

Churn Analysis at Fit.ly Tech

Prepared for: Senior Leadership Team

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Project Overview

Over the past two quarters, Fit.ly Tech has experienced a noticeable increase in subscriber churn. As customer acquisition costs continue to rise, retaining existing subscribers has become a critical business priority.

Every customer who leaves:

- Reduces recurring subscription revenue
 - Increases pressure on Marketing to acquire new users
 - Impacts long-term business growth and profitability
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Business Goals

This analysis was conducted to help Fit.ly Tech:

- ✓ Reduce customer churn
- ✓ Improve subscriber retention
- ✓ Increase customer lifetime value
- ✓ Enhance customer engagement
- ✓ Improve customer experience

Key Business Questions

1. Which customer segments are most likely to churn?
2. How does customer engagement influence retention?
3. Does customer support activity contribute to churn?
4. Are certain subscription plans associated with higher churn rates?
5. Which KPIs should leadership monitor to proactively reduce churn?

Data validation and cleaning

The data consisted of three source tables:

- **account_info:** 400 rows, 6 columns
- **customer_support:** 918 rows, 7 columns
- **user_activity:** 445 rows, 3 columns

account_info (400, 6)

- (i) customer_id

Checked as the unique customer identifier. The non-numeric portion was removed, converted to int64, and renamed to user_id to facilitate joins across datasets.

- (ii) Email

Retained as a customer identifier but not used as a potential churn predictor.

- (iii) State

Renamed to customer_state to avoid confusion with the state field in the support ticket dataset. There were 50 unique customer_state

- (iv) plan

Unique values are Basic, Free, Enterprise and Pro. No value inconsistencies.

- (v) plan_list_price

Already stored as int64 and contained no outliers. Minimum price 0 for free users and maximum 148 for paid users which is below the upper band of 193.125.

- (vi) churn_status

Renamed to churn and converted to a binary variable where Y = 1 (churned) and all missing values were set to 0 (retained)

customer_support (918, 7) → (367, 7)

(i) ticket_time

Parsed and converted to a datetime format.

(ii) user_id

Already stored as int64 and used as the customer identifier.

(iii) channel

There were 3 unique entries: email, chat and phone. While (-) was converted to unknown.

(iv) topic

There were 4 unique entries: billing, account, technical and other. support ticket counts was created by grouping up the user_id and taking count for each user.

(v) resolution_time_hours

Summarized by grouping up user_id to create average and maximum resolution times columns.

(vi) state

Renamed to ticket_closed. An unresolved_tickets indicator was also created to capture unresolved support cases where values were 0

(vii) comments

Missing comments were filled with blank text. Additionally, a has_delete_request flag was created to identify comments containing data deletion requests.

After aggregation to the customer level, the dataset was reduced from **918 rows to 367 unique customers**.

user_activity (445, 3) → (246, 5)

(i) event_time

Parsed to datetime format, grouped by user_id and created new columns namely, number of active days, First event date and Last event date

(ii) user_id

Already stored as int64 and used as the customer identifier.

(iii) event_type

There were 4 unique entries: read_article, watch_video, track_workout and share_workout.

Grouped by user_id and created a new column, activity_events count for each user. After aggregation, the dataset was reduced from **445 rows to 246 unique customers**.

No exact **duplicate rows** across any of the datasets. However, duplicate **user_id** values existed because customers could have multiple support tickets and activity events. To create a customer-level dataset, records were grouped by **user_id** and summarized into activity and support metrics, resulting in a unique record for each customer.

Missing values only existed in the **account_info (churn_status)** and was treated as 0 for not churned and also in **customer_support (comments.)** which was converted to blank.

Final Customer Dataset (400, 16)

The activity and support summaries were **left-joined** to the **account_info** table to create the final customer dataset.

Customers with no recorded activity or support interactions were retained and their summary metrics were filled with zeros. This approach avoids excluding low-engagement customers who may be particularly important for churn analysis.

The final dataset contains **400 unique customers and 16 features**.

Business metric to monitor

Which KPIs should leadership monitor to proactively reduce churn?

Recommended KPIs for a **Retention Dashboard**:

- **Overall retention rate** (currently 71.5%).
- **Churn rate by plan** (to spot plan-specific risks).
- **Churn rate by engagement band** (low vs. high activity).
- **Support ticket volume** (especially unresolved tickets).
- **Delete request rate** (critical churn signal).

These KPIs provide early warning signals and allow leadership to act before churn escalates.

The main metric is customer retention rate:

retention rate = retained customers / total customers

This is the best business metric because the problem is churn, and retention directly measures whether Fit.ly is keeping subscribers.

It can be tracked monthly or quarterly by plan, and connects to revenue pressure because retained customers continue paying without new acquisition cost.

Current baseline from this dataset:

Customers: 400

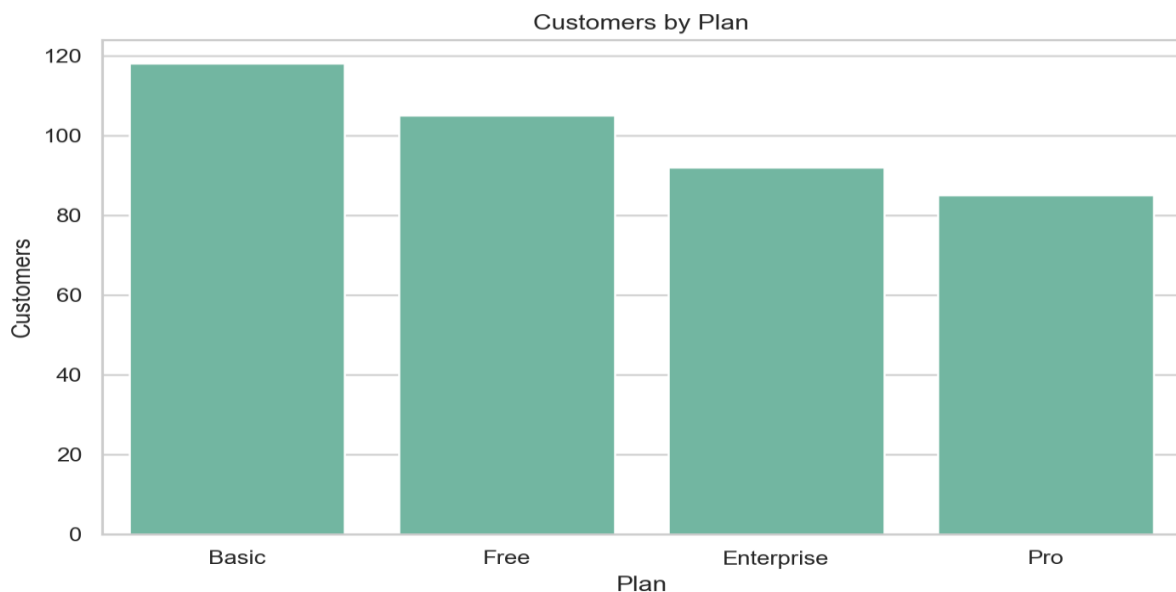
Churned customers: 114

Retention rate: 71.5%

Churn rate: 28.5%

Leadership should track retention overall and by plan, with a supporting operational target to reduce churn among customers with high support volume or deleted requests.

Customers By Subscription Plan



This bar chart shows the number of customers in each plan category. The visual compares the size of Fit.ly's customer base across Free, Basic, Pro, and Enterprise plans.

From the chart, the bars are fairly close in height, but Basic has the largest customer group, followed by Free, then Enterprise, while Pro has the smallest group. This means Fit.ly's customer base is spread across the plans, but Basic and Free users make up a large share of the total customer base.

This gives important business context. If a plan has both a high customer count and high churn, it will have a larger effect on Fit.ly's overall retention. This is why plan size matters in addressing churn.

Fit.ly should monitor retention by plan because losses in larger plan groups, especially Basic and Free, can strongly affect total customer retention.

Distribution Of Customer Activity Events



This histogram shows the distribution of activity events per customer.

The x-axis represents the number of activity events recorded by a customer, while the y-axis shows how many customers fall into each activity level.

The visual suggests that many customers have low activity counts because distribution is concentrated toward the lower end, meaning a large portion of users have only a few recorded events.

How does customer engagement influence retention?

- Average engagement is **1.11 activity events per customer**, which is very low.
- Customers who do not interact much with the app may not see enough value to continue using Fit.ly.

Fit.ly should improve onboarding, reminders, personalized workout suggestions, and habit-building features to increase app engagement.

Low activity is connected to churn risk.

Key findings

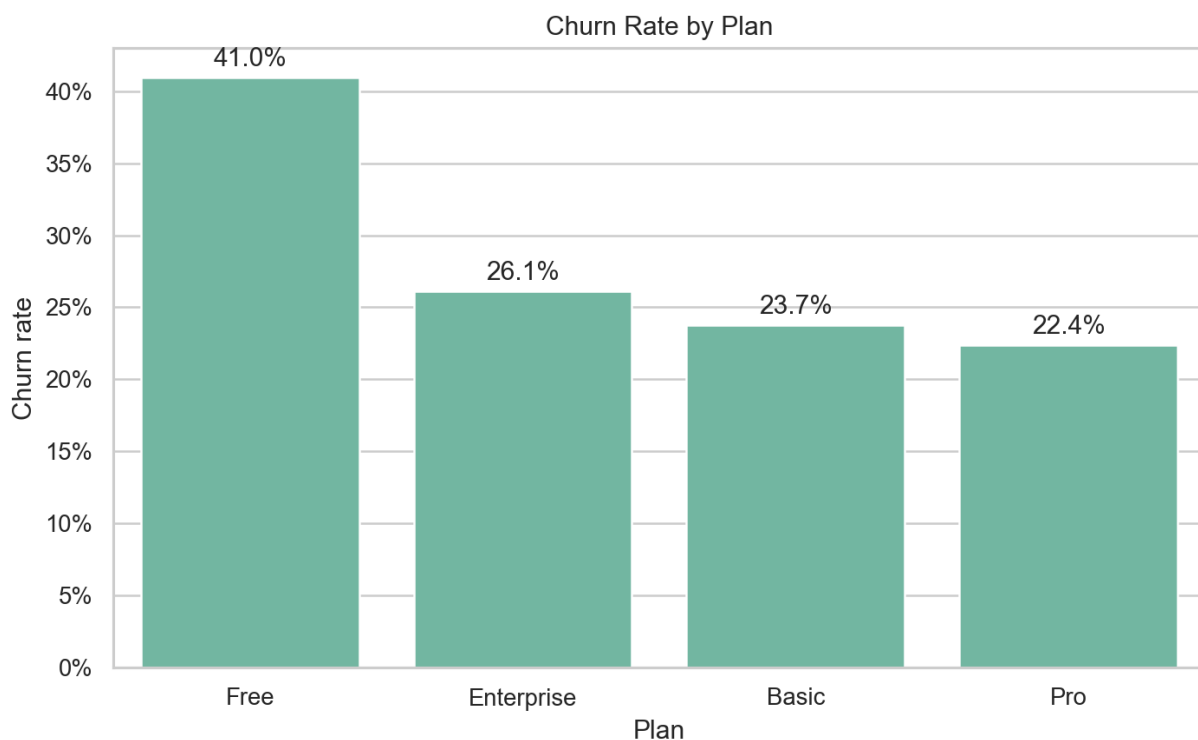
Average customer engagement is 1.11 activity events per customer.

Engagement should be monitored because customers with little or no product activity are less likely to be using the app.

Average support volume is 2.29 tickets per customer.

Support interactions matter because repeated issues and slow resolution can create churn risk. Customers with delete/account/data removal requests have a churn rate of 45.5%. This is a strong operational warning signal and should be routed for immediate retention follow-up.

Churn Rate By Plan



This bar chart compares churn rate across each plan. The height of each bar shows the percentage of customers who churned in that plan.

Which customer segments are most likely to churn?

- **Free plan users** have the highest churn rate at **41.0%**.
- **Enterprise plan** churn is also relatively high at **26.1%**, while **Pro plan** has the lowest churn at **22.4%**.
- Customers with **delete/account removal requests** show a churn rate of **45.5%**.

High-risk segments: Free users, customers submitting delete requests, and those with unresolved support issues.

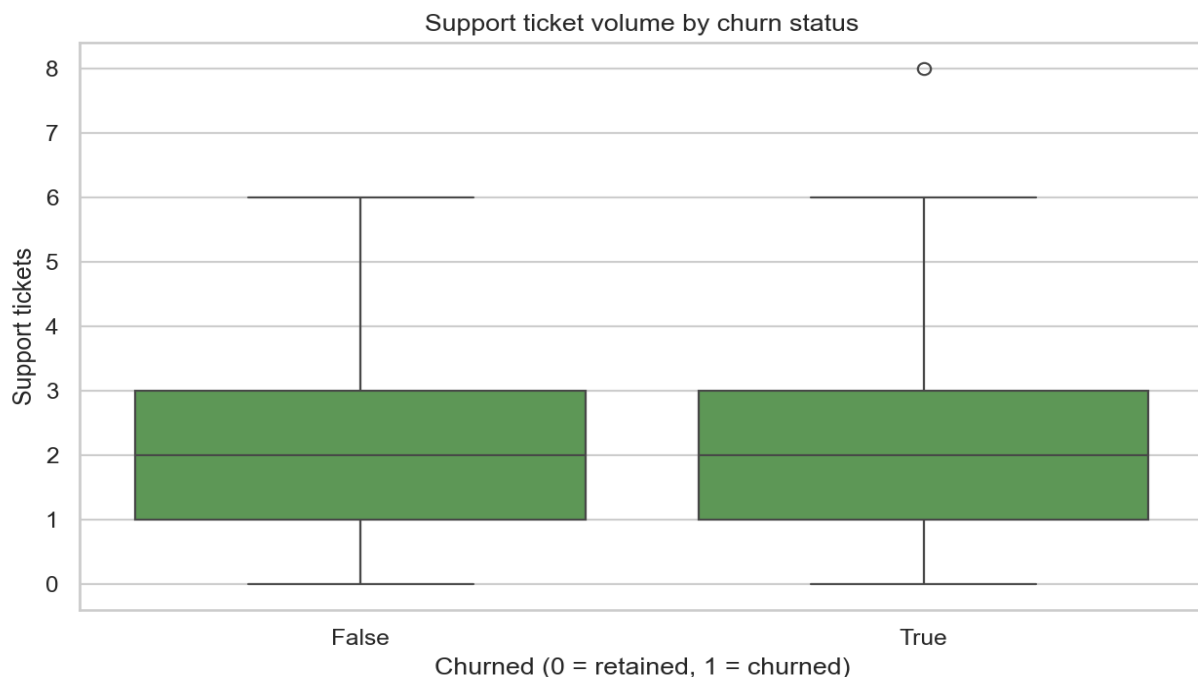
This suggests pricing or value perception issues that leadership should investigate.”

Free users are the most likely to leave. This may be because Free users have lower commitment, less access to premium value, or weaker motivation to stay.

Pro users have the lowest churn, which may suggest stronger product fit or greater perceived value among paying users.

Fit.ly should focus on improving Free-user onboarding and creating stronger upgrade or engagement pathways.

Support Ticket Volume By Churn Status



Support activity is another churn driver. Customers with more tickets are more likely to churn, especially if tickets remain unresolved. Most importantly, customers who request account or data deletion have a very high churn rate which is a clear warning signal.

This chart compares `support\_tickets` between customers who were retained and customers who churned. The churn status variable separates customers into two groups: `0 = retained` and `1 = churned`.

The visual communicates whether churned customers tend to have more support interactions than retained customers. If the churned group shows a higher support ticket level, it suggests that customers who experience more issues or need more help are more likely to leave. Support tickets act as a signal of customer friction.

Does customer support activity contribute to churn?

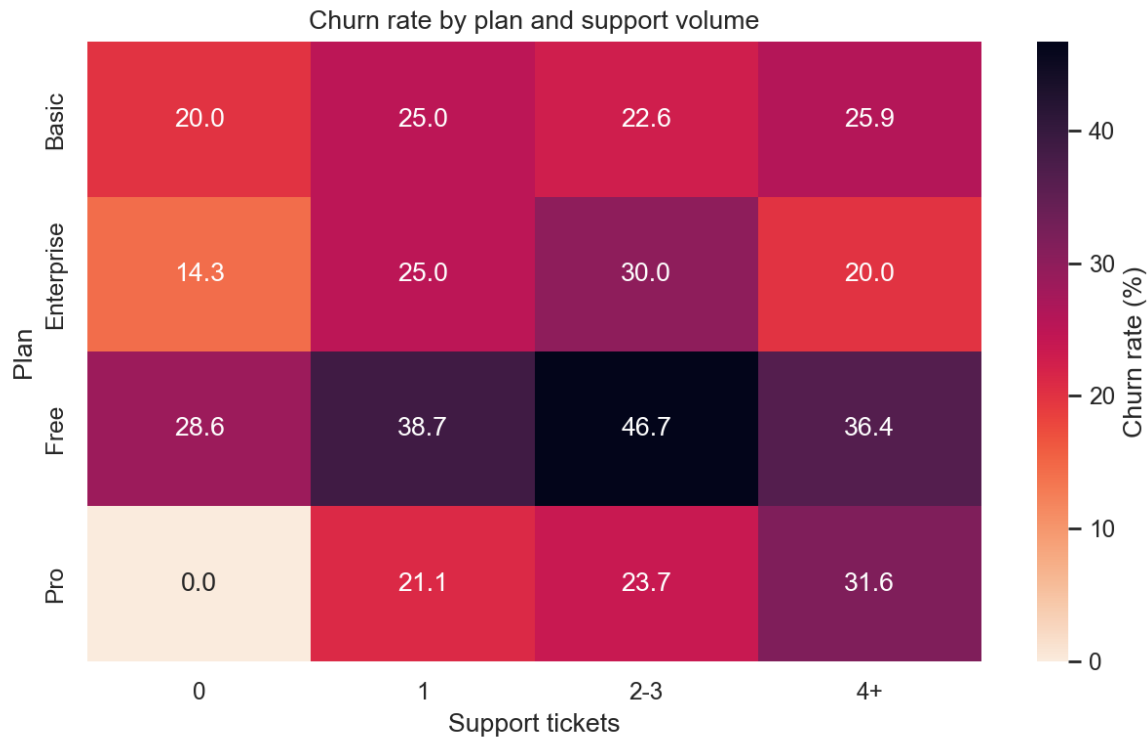
- Customers with **higher support ticket volumes** are more likely to churn.
- **Unresolved tickets** and **slow resolution times** increase churn risk.
- **Delete requests** are the strongest churn signal (45.5% churn).

Support activity is a leading indicator of churn. High-ticket customers should be prioritized for fast resolution and proactive outreach.

Customers who contact support repeatedly may be experiencing unresolved problems, confusion, billing concerns, or technical barriers. These issues can reduce satisfaction and increase churn risk.

Fit.ly should treat high support activity as an early warning signal. Customers with multiple tickets should be prioritized for faster resolution or proactive retention follow-up.

Churn Rate By Plan And Support Volume



This heatmap shows the relationship between three variables: subscription plan, support tickets, and churn rate.

Each cell represents the churn rate for customers in a specific plan and support-ticket group.

The visual uses color intensity to communicate risk. Darker or more intense cells represent higher churn rates, while lighter cells represent lower churn rates. This makes it easier to identify which combinations of plan and support activity have the highest churn risk.

This helps reveal whether support issues affect churn differently across plans.

Customers are not equally risky just because they belong to the same plan.

A Free customer with multiple support tickets may have a different churn risk than a Pro customer with no support tickets.

The chart helps show that churn risk may increase when plan type and customer friction happen together.

Fit.ly should use both plan and support activity to prioritize retention efforts.

The highest-risk groups should receive proactive support, faster issue resolution, and targeted engagement campaigns.

To summarize:

churn is uneven across plans, low engagement customers are at risk, high support volume correlates with churn, and delete requests are a strong churn signal.

Recommended actions

My recommendations are:

To build a retention dashboard that tracks overall retention rate, churn rate by plan, churn rate by engagement band, support ticket volume, and delete request rate.

Prioritize outreach to customers with delete requests, unresolved tickets, or multiple recent tickets. These are explicit signs of friction.

Review onboarding and in-product nudges for low-activity customers, especially those with zero or one activity event.

Compare pricing and value perception for the highest-churn plan. If churn is concentrated there, leadership should test plan-specific retention offers or product improvements.

Improve support workflows for common high-volume topics and measure whether faster resolution reduces churn in the next quarter.